

HDOC Day-37

String Manipulation Using a Menu-Driven C Program

Aim

To write a menu-driven C program using separate functions to convert lowercase text to uppercase without built-in conversion functions, reverse a string, check whether a string is a palindrome, and exit.

Algorithm

1. Start the program and read a string.
2. Display the four menu options.
3. Accept the user's choice and use a switch statement.
4. For uppercase conversion, change each lowercase letter by subtracting 32 from its ASCII value.
5. For reversal, find the length manually and print from the last character to the first.
6. For palindrome checking, compare matching characters from both ends.
7. Exit when option 4 is selected.
8. Display an error for invalid choices and repeat the menu until Exit.

C Program

```
#include <stdio.h>

void toUppercase(char str[]);
void reverseString(char str[]);
void checkPalindrome(char str[]);

int main()
{
    char str[200];
    int choice, i = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do
    {
        printf("\n----- STRING OPERATIONS MENU ----- \n");
        printf("1. Convert Lowercase String to Uppercase\n");
        printf("2. Reverse a String\n");
        printf("3. Check Palindrome\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: toUppercase(str); break;
            case 2: reverseString(str); break;
            case 3: checkPalindrome(str); break;
            case 4: printf("Exiting program...\n"); break;
        }
    }
}
```

```

        default:
            printf("Invalid choice! Please enter a number from 1 to 4.\n");
        }
    } while (choice != 4);

    return 0;
}

void toUppercase(char str[])
{
    int i = 0;
    char upper[200];

    while (str[i] != '\0')
    {
        if (str[i] >= 'a' && str[i] <= 'z')
            upper[i] = str[i] - 32;
        else
            upper[i] = str[i];
        i++;
    }
    upper[i] = '\0';
    printf("Uppercase String: %s\n", upper);
}

void reverseString(char str[])
{
    int length = 0, i;
    while (str[length] != '\0')
        length++;

    printf("Reversed String: ");
    for (i = length - 1; i >= 0; i--)
        printf("%c", str[i]);
    printf("\n");
}

void checkPalindrome(char str[])
{
    int length = 0, i, flag = 1;
    while (str[length] != '\0')
        length++;

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - i - 1])
        {
            flag = 0;
            break;
        }
    }

    if (flag)
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");
}

```

Test Case Table

1. Uppercase: hello -> HELLO
2. Uppercase: computer science -> COMPUTER SCIENCE
3. Reverse: hello -> olleh
4. Reverse: UPES -> SEPU
5. Palindrome: madam -> palindrome
6. Palindrome: level -> palindrome
7. Palindrome: hello -> not a palindrome

8. Invalid choice: 5 -> error message

9. Exit: 4 -> program exits

Sample Output

```
Enter a string: madam
Enter your choice: 1
Uppercase String: MADAM
Enter your choice: 2
Reversed String: madam
Enter your choice: 3
The string is a palindrome.
Enter your choice: 4
Exiting program...
```

Result

The menu-driven C program was successfully implemented using separate functions, strings, loops, conditional statements, and switch-case. It performs uppercase conversion without built-in conversion functions, reverses strings, checks palindromes, and exits through the menu.